

IGL VisionGuard AI — System Workflow Report

Complete Technical Architecture, AI Pipeline & Module Walkthrough Report

1. Executive Summary

IGL VisionGuard AI is an enterprise-grade real-time computer vision platform designed for industrial safety compliance and high-speed conveyor packaging inspection. It integrates real-time person tracking, PPE helmet verification, two-pass mobile phone misuse escalation, proximity radar, and automated 3D bottle quality control.

2. AI Vision Pipeline Architecture

Two-Pass Mobile Phone Detection:

- *Pass 1 (Full Frame)*: Executes COCO-SSD detection at score threshold 0.15.
- *Pass 2 (Upper Body ROI)*: Crops person upper-body, upscales to 360x360, re-inspects for small phone candidates.
- *IoU Deduplication*: Merges overlapping detections (>0.30 IoU) into a unified prediction list.

Dedicated ML Helmet Model:

- Head crop ROI tensor processing (224x224 input).
- Conservative default setting (hasHelmet=false) when custom model files are unsupplied.
- Temporal memory (3 persistence frames + 3 miss grace frames).

3. Application Sitemap & Module Overview

#	Module Name	Primary Technical Function	Key Components
1	Overview	High-level KPI stats & live alert stream	DashboardView.jsx
2	Camera Setup	Webcam, RTSP & MP4 input + Mirror toggle	CameraInputView.jsx
3	Live Monitoring	60 FPS canvas AI detection + Siren state machine	LiveMonitoringView.jsx
4	Packaging Inspection	Bottle Fill-level, Cap integrity & Body defect QC	PackagingInspectionView.jsx
5	Safety Events	Alert incident log with base64 photo evidence	AlertsView.jsx
6	Near Miss	Dynamic distance radar & 2.0m Safety Boundary	NearMissView.jsx
7	Incidents	Personnel recognition & Sleeping-on-duty alerts	PersonnelAnomalyView.jsx
8	Analytics	Historical safety breach trends & charts	AnalyticsView.jsx
9	Camera Health	Lens blur variance & brightness quality scoring	CameraHealthView.jsx
10	Configuration	Geofence dangerous zone polygon builder	ConfigurationView.jsx

4. Backend Database & REST API Contracts

FastAPI endpoints connected to SQLite database igl_safety.db:

Method	Endpoint	Description
GET	/api/dashboard/stats	Calculates live stats directly from SQLite tables
GET	/api/cameras	Retrieves active registered camera channels
POST	/api/alerts	Saves safety violation with base64 evidence snapshot
PATCH	/api/alerts/{id}/status	Updates alert status to ACKNOWLEDGED or RESOLVED